

Advanced Mathematical Techniques

Problem Set 7

1.

$$\begin{aligned} \int_0^1 \frac{\ln^2(1-t^2)}{t^4} dt &\xrightarrow{u=t^2} \int_0^1 \frac{\ln^2(1-u)}{u^2} \frac{1}{2u^{1/2}} du = \frac{1}{2} \int_0^1 (1-u)^\varepsilon u^{-5/2+\delta} du = \frac{\Gamma(1+\varepsilon)\Gamma(-3/2+\delta)}{2\Gamma(-1/2+\varepsilon+\delta)} \\ &= \frac{\Gamma(1+\varepsilon)}{2} \cdot \frac{\Gamma(-1/2+\delta)}{(-3/2+\delta)(-1/2+\delta)\Gamma(1/2+\varepsilon+\delta)} \xrightarrow{\delta=0} \frac{\Gamma(1+\varepsilon) \cdot \frac{4}{3}\sqrt{\pi}}{2 \cdot \frac{3}{4}\sqrt{\pi}\Gamma(1/2+\varepsilon)} \\ &= \frac{e^{(2\varepsilon+1)\ln 2}\Gamma(1+\varepsilon)\Gamma(1+\varepsilon)}{3\Gamma(1+2\varepsilon)} = \frac{1}{3}(\varepsilon-1/2) \exp\{\ln 2 + 2\varepsilon \ln 2 - \gamma(2\varepsilon-2\varepsilon) + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} (2-2^k)\varepsilon^k\} \end{aligned}$$

We only need the ε^2 coefficient, so we consider only

$$\frac{4}{3} \left(2\ln 2 - \frac{1}{2} \left(2\ln^2 2 + \frac{\zeta(2)}{2} (-1/2) \right) \right) = \boxed{\frac{\pi^2}{9} + \frac{8}{3} \ln 2 - \frac{4}{3} \ln^2 2}$$

2. $\int_0^1 \frac{\ln^2(1-t)}{t^3 \sqrt{t}} dt$ diverges, since at 0, $\ln(1-t) \approx -t$, so the integral simplifies to $\int_0^1 \frac{-1}{t\sqrt{t}}$, which diverges at 0.

3.

$$\begin{aligned} \int_0^1 \frac{\ln^2(1-t^2) \ln t}{t^3} dt &\xrightarrow{u=t^2} \frac{1}{4} \int_0^1 \frac{\ln^2(1-u) \ln u}{2u^2} du \rightarrow \int_0^1 u^{-2+\delta} (1-u)^\varepsilon = \frac{\Gamma(1+\varepsilon)\Gamma(1+\delta)(\varepsilon+\delta)}{\Gamma(1+\varepsilon+\delta)(1+\delta)(\delta)} \\ &= \frac{(\varepsilon+\delta)}{\delta(1+\delta)} \exp\{-\gamma(1-\delta+\varepsilon-\delta) + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} [\varepsilon^k + \delta^k - (\varepsilon+\delta)^k]\} = \frac{1}{\delta} \cdot (\varepsilon+\delta) \cdot (1+\delta+\delta^2+\dots) \cdot \exp\{\text{previous}\} \end{aligned}$$

We only need to consider the $\varepsilon\delta$ coefficient, so

$$-\frac{1}{2} \left(\frac{\zeta(2)}{2} \right) (-\delta) + \frac{\zeta(3)}{3} (-3) = \boxed{\frac{\pi^2}{12} - \zeta(3)}$$

4.

$$\int_0^\infty \frac{1-e^{-3x}}{x^{\frac{3}{2}}} dx \rightarrow \int_0^\infty x^{-3/2+\delta} (1-e^{-3x}) dx = \int_0^\infty x^{-3/2+\delta} dx - \int_0^\infty x^{-3/2+\delta} e^{-3x} dx = [x^{-1/2+\delta}]_0^\infty - \frac{\Gamma(-1/2+\delta)}{3^{-1/2+\delta}}$$

The first term is useless, so let's ignore it.

$$= -3e^{-\delta \ln 3} 2 \cdot 2^{-2\delta} \sqrt{\pi} \frac{\Gamma(1+2\delta)}{(1-2\delta)\Gamma(1+\delta)} = -2\sqrt{3\pi} (1+2\delta+4\delta^2+\dots) \exp\{-\gamma\delta - \delta \ln 3 - 2\delta \ln 2 + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} (2^k-1)\delta^k\}$$

Since the δ^0 coefficient is $2\sqrt{3\pi}$, the integral is $2\sqrt{3\pi}$.

5.

$$\begin{aligned} \int_0^\infty \frac{x}{e^{3x}-1} dx &= \int_0^\infty \frac{x e^{-3x}}{1-e^{-3x}} dx \xrightarrow{u=3x} \frac{1}{9} \int_0^\infty \frac{u e^{-u}}{1-e^{-u}} du = \frac{1}{9} \sum_{k=0}^{\infty} \int_0^\infty u e^{-u} e^{-uk} du \\ &= \frac{1}{9} \sum_{k=0}^{\infty} \int_0^\infty u e^{-(k+1)u} du = \frac{1}{9} \sum_{k=0}^{\infty} \frac{\Gamma(2)}{(k+1)^2} = \frac{1}{9} \zeta(2) \Gamma(2) = \boxed{\frac{\pi^2}{54}} \end{aligned}$$

6.

$$\int_0^\infty \frac{x^2 e^{2x}}{(e^{2x} - 1)^2} dx = -\frac{\partial}{\partial a} \Big|_{a=2} \int_0^\infty \frac{x}{e^{ax} - 1} dx = -\frac{\partial}{\partial a} \Big|_{a=2} \frac{\Gamma(2)\zeta(2)}{a^2} = \frac{2\zeta(2)}{2^3} = \frac{\pi^2}{3 \cdot 2^3} = \boxed{\frac{\pi^2}{24}}$$

7.

$$\begin{aligned} \int_0^\infty \frac{x^3}{(e^{2x} - 1)^2} dx &= -\frac{\partial}{\partial a} \Big|_{a=2} \int_0^\infty \frac{x^2}{e^{ax} - 1} dx - \int_0^\infty \frac{x^3}{e^{ax} - 1} dx = -\frac{\partial}{\partial a} \Big|_{a=2} \frac{\Gamma(3)\zeta(3)}{a^3} - \frac{\Gamma(4)\zeta(4)}{a^4} = \frac{6\zeta(3)}{a^4} - \frac{6\zeta(4)}{a^4} \\ &= \frac{6}{16} \left(\zeta(3) - \frac{\pi^4}{90} \right) = \boxed{\frac{3}{8}\zeta(3) - \frac{\pi^4}{240}} \end{aligned}$$

8.

$$\int_0^1 \frac{\ln^2(1-t^2) \ln t}{t^2} dt = \int_0^1 t^{-2+\delta} (1-t)^\varepsilon dt = \frac{\Gamma(1+\varepsilon)\Gamma(-1+\delta)}{\Gamma(\varepsilon+\delta)}$$

which is the same Beta function integral as #3, so the $\varepsilon\delta$ coefficient is still $\frac{\pi^2}{6} - 2\zeta(3)$.

$$\therefore \int_0^1 \frac{\ln^2(1-t^2) \ln t}{t^2} dt = 2! \cdot 1! \cdot \left(\frac{\pi^2}{6} - 2\zeta(3) \right) = \boxed{\frac{\pi^2}{3} - 4\zeta(3)}$$